

SALES AND PROFITABILITY ANALYSIS

Author: Lamidi Khadijat

Tool Used: Power BI

BUSINESS OVERVIEW

The company operating across six countries and ten regions sells bikes, clothing, and accessories. While sales volume appeared to increase over time, revenue growth did not always rise at the same rate, making it difficult to identify the factors influencing business performance.

PROBLEM STATEMENT

Despite strong sales activity across products and regions, it is unclear which factors contribute most to revenue and profitability. Increased sales volume does not always lead to proportional revenue or profit growth, making it important to understand the impact of product mix, customer behavior, and regional performance.

BUSINESS QUESTIONS:

This is sectioned into three parts :

- A. Executive Summary
 - How much revenue, cost, and profit are generated?
 - Which product categories contribute the most to total revenue?
 - Which countries and regions drive performance?
 - What are the overall sales trends?
- B. Product & Profitability Analysis
 - Which product Category generate the highest revenue?
 - Which products Model Name generate the highest Revenue?
 - Are the best-selling products also the most profitable products?
 - Is the current product mix maximizing profitability?
- C. Customer Behavior Analysis
 - Who are the highest-value customers based on purchasing behavior?
 - How does customer purchasing behavior change over time?
 - Which sales territories have strong sales but weak profitability?
 - Is there variation in gender contribution to revenue?
 - Which age groups contribute most to revenue?

DATA ANALYSIS EXPRESSIONS USED

1. Total Revenue

Total Revenue = SUM('Sales'[Total price])

2. Total Revenue without filter context

Total_rev = CALCULATE([Total Revenue],ALL('Sales'[Total price]))

Total Revenue Vs Previous Month =

Total Revenue vs PM =

```
VAR CurrentMonth = [Total Revenue]
VAR LastMonth =
    CALCULATE(
        [Total Revenue], DATEADD('Date'[Date], -1, MONTH)
    )
RETURN
    IF(
        ISBLANK(LastMonth),
        BLANK(),
        CurrentMonth - LastMonth
    )
```

3. Total Revenue MoM label

Total revenue MoM label =

```
VAR _variance = [Total Revenue vs PM]
RETURN
    IF(ISBLANK(_variance),
        "vs Previous Month: ---",
        VAR _absVariance = ABS(_variance)
        VAR ICON =
            IF(_variance = 0, " ",
            IF(_variance > 0, "▲", "▼"))
        VAR FORMATTEDVALUE =
            SWITCH(
                TRUE(),
                _absVariance >= 1000000,
                FORMAT(_absVariance/1000000, "$#,##0.00\M"),
                _absVariance >= 1000, FORMAT(_absVariance / 1000, "$#,##0.00\K"),
                FORMAT(_absVariance, "$#,##0")
            )
        "vs Precious Month: "&ICON & " " & FormattedValue
    )
```

4. Revenue Label Color

Label color =

```
IF([Total Revenue vs PM] = 0, "Black",  
IF([Total Revenue vs PM] > 0, "Green", "Red"))
```

5. Total Cost

Total Cost = `SUM('Sales'[Total cost])`

6. Total Cost Vs Previous Month

Total Revenue vs PM =

```
VAR CurrentMonth = [Total Cost]  
VAR LastMonth =  
    CALCULATE(  
        [Total Cost], DATEADD('Date'[Date], -1, MONTH)  
    )  
RETURN  
IF(  
    ISBLANK(LastMonth),  
    BLANK(),  
    CurrentMonth - LastMonth  
)
```

7. Total Cost MoM label

Total Cost MoM label =

```
VAR _variance = [Total Cost vs PM]  
RETURN  
IF(ISBLANK(_variance),  
"vs Previous Month: ---",  
VAR _absVariance = ABS(_variance)  
VAR ICON =  
    IF(_variance = 0, " ",  
    IF(_variance > 0, "▲", "▼"))  
VAR FORMATTEDVALUE =  
    SWITCH(  
        TRUE(),  
        _absVariance >= 1000000,  
        FORMAT(_absVariance/1000000, "$#,##0.00\M"),  
        _absVariance >= 1000, FORMAT(_absVariance / 1000, "$#,##0.00\K"),  
        FORMAT(_absVariance, "$#,##0")
```

```
)
RETURN
"vs Precious Month: "&ICON & " " & FormattedValue
)
```

8. Cost Label Color

```
Cost MoM Label =
IF([Total Cost vs PM] = 0, "Black",
IF([Total Cost vs PM] > 0, "Green", "Red"))
```

9. Total Profit

```
Total Profit = SUM([Total Revenue])
```

10. Total Profit Vs Previous Month =

```
Total Profit vs PM =
```

```
VAR CurrentMonth = [Total Profit]
VAR LastMonth =
    CALCULATE(
        [Total Profit], DATEADD('Date'[Date], -1, MONTH)
    )
RETURN
IF(
    ISBLANK(LastMonth),
    BLANK(),
    CurrentMonth - LastMonth
)
```

11. Total Profit MoM label

```
Total Profit MoM label =
```

```
VAR _variance = [Total Profit vs PM]
RETURN
IF(ISBLANK(_variance),
"vs Previous Month: ---",
VAR _absVariance = ABS(_variance)
```

```

VAR ICON =
    IF(_variance = 0, " ",
    IF(_variance > 0, "▲", "▼"))
VAR FORMATTEDVALUE =
    SWITCH(
        TRUE(),
        _absVariance >= 1000000,
        FORMAT(_absVariance/1000000, "$#,##0.00\M"),
        _absVariance >= 1000, FORMAT(_absVariance / 1000, "$#,##0.00\K"),
        FORMAT(_absVariance, "$#,##0")
    )
RETURN
"vs Precious Month: "&ICON & " " & FormattedValue
)

```

12. Profit Label Color

Profit Label color =

```

IF([Total Profit vs PM] = 0, "Black",
IF([Total Profit vs PM] > 0, "Green", "Red"))

```

13. Margin%

Margin% = `DIVIDE(SUM('Sales'[Profit]),SUM('Sales'[Total price]))`

14. Margin Vs Previous Month

Margin vs PM =

```

VAR CurrentMonth = [Margin%]
VAR LastMonth =
    CALCULATE(
        [Margin%], DATEADD('Date'[Date], -1, MONTH)
    )
RETURN
IF(
    ISBLANK(LastMonth),
    BLANK(),
    CurrentMonth - LastMonth
)

```

15. Total Margin MoM label

Margin MoM label =

```

VAR _variance = [Margin vs PM]
RETURN
IF(ISBLANK(_variance),
"vs Previous Month: ---",

VAR ICON =
    IF(_variance >= 0, "▲", "▼")
RETURN
"vs Precious Month: " & ICON & " " & FORMAT(ABS(_variance), "0.00%")
)

```

16. Margin Label Color

```

Margin label color =
IF([Margin vs PM] = 0, "Black",
IF([Margin vs PM] > 0.40, "Green", "Red"))

```

17. Sales Volume

```

Sales Volume = COUNT('Sales'[Price])

```

Sales Volume Vs Previous Month

sales volume vs PM =

```

VAR CurrentMonth = [Sales Volume]
VAR LastMonth =
    CALCULATE(
        [Sales Volume], DATEADD('Date'[Date], -1, MONTH)
    )
RETURN
IF(
    ISBLANK(LastMonth),
    BLANK(),
    CurrentMonth - LastMonth
)

```

18. Sales Volume MoM label

sales volume MoM label =

```

VAR _variance = [sales volume vs PM]
RETURN
IF(ISBLANK(_variance),
"vs Previous Month: ---",
VAR _absVariance = ABS(_variance)
)

```

```

VAR ICON =
    IF(_variance = 0, " ",
    IF(_variance > 0, "▲", "▼"))
VAR FORMATTEDVALUE =
    SWITCH(
        TRUE(),
        _absVariance >= 1000000,
        FORMAT(_absVariance/1000000, "#,##0.00\M"),
        _absVariance >= 1000, FORMAT(_absVariance / 1000, "#,##0.00\K"),
        FORMAT(_absVariance, "#,##0")
    )
RETURN
"vs Precious Month: "&ICON & " " & FormattedValue
)

```

19. Sales Volume Label Color

```

sales volume label =
    IF([sales volume vs PM] = 0, "Black",
    IF([sales volume vs PM] > 0, "Green", "Red"))

```

20. Total Quantity

Total Quantity = SUM('Sales'[OrderQuantity])

21. Total Quantity Vs Previous Month

Total Quantity vs PM =

```

VAR CurrentMonth = [Total Quantity]
VAR LastMonth =
    CALCULATE(
        [Total Quantity], DATEADD('Date'[Date], -1, MONTH)
    )
RETURN
    IF(
        ISBLANK(LastMonth),
        BLANK(),
        CurrentMonth - LastMonth
    )

```

22. Total Quantity MoM label

Total quantity MoM label =

```
VAR _variance = [Total Quantity vs PM]
RETURN
IF(ISBLANK(_variance),
"vs Previous Month: ---",
VAR _absVariance = ABS(_variance)
VAR ICON =
    IF(_variance = 0, " ",
    IF(_variance > 0, "▲", "▼"))
VAR FORMATTEDVALUE =
    SWITCH(
        TRUE(),
        _absVariance >= 1000000,
        FORMAT(_absVariance/1000000, "#,##0.00\M"),
        _absVariance >= 1000, FORMAT(_absVariance / 1000, "#,##0.00\K"),
        FORMAT(_absVariance, "#,##0")
    )
RETURN
"vs Precious Month: "&ICON & " " & FormattedValue
)
```

23. Total Quantity label

Q_Label =

```
IF([Total Quantity vs PM] = 0, "Black",
IF([Total Quantity vs PM] > 0, "Green", "Red"))
```

24. Customer Count

Customer Count = `DISTINCTCOUNT('Sales'[CustomerKey])`

Customer Count Vs Previous Month

Customers vs PM =

```
VAR CurrentMonth = [Customer Count]
VAR LastMonth =
    CALCULATE(
        [Customer Count], DATEADD('Date'[Date], -1, MONTH)
    )
```



```

RETURN
IF(
    ISBLANK(LastMonth),
    BLANK(),
    CurrentMonth - LastMonth
)

```

25. Customer Count MoM Label

Customer MoM label =

```

VAR _variance = [Customers vs PM]
RETURN
IF(ISBLANK(_variance),
    "vs Previous Month: ---",
    VAR _absVariance = ABS(_variance)
    VAR ICON =
        IF(_variance = 0, " ",
            IF(_variance > 0, "▲", "▼"))
    VAR FORMATTEDVALUE =
        SWITCH(
            TRUE(),
            _absVariance >= 1000000,
            FORMAT(_absVariance/1000000, "#,##0.00\M"),
            _absVariance >= 1000, FORMAT(_absVariance / 1000, "#,##0.00\K"),
            FORMAT(_absVariance, "#,##0")
        )
    )
RETURN
"vs Precious Month: "&ICON & " " & FormattedValue
)

```

26. Customer Count PM Label

Customer_label =

```

IF([Customers vs PM] = 0, "Black",
    IF([Customers vs PM] > 0, "Green", "Red"))

```

27. Products below Margin Target

Products Below Margin =

```

COUNTX(VALUES('Sales'[ProductKey]),
    IF([Margin%] < 0.40, 1))

```

28. Products Below Vs Previous Month

Products Below vs PM =

```
VAR CurrentMonth = [Products Below Margin]
VAR LastMonth =
    CALCULATE(
        [Products Below Margin], DATEADD('Date'[Date], -1, MONTH)
    )
RETURN
    IF(
        ISBLANK(LastMonth),
        BLANK(),
        CurrentMonth - LastMonth
    )
```

29. Products Below MoM label

products below MoM label =

```
VAR _variance = [Products Below vs PM]
RETURN
    IF(ISBLANK(_variance),
        "vs Previous Month: ---",
        VAR _absVariance = ABS(_variance)
        VAR ICON =
            IF(_variance = 0, " ",
            IF(_variance > 0, "▲", "▼"))
        VAR FORMATTEDVALUE =
            SWITCH(
                TRUE(),
                _absVariance >= 1000000,
                FORMAT(_absVariance/1000000, "#,##0.00\M"),
                _absVariance >= 1000, FORMAT(_absVariance / 1000, "#,##0.00\K"),
                FORMAT(_absVariance, "#,##0")
            )
    )
RETURN
    "vs Precious Month: "&ICON & " " & FormattedValue
)
```

30. Products Below Label Color

products below label =

```
IF([Products Below vs PM] = 0, "Black",
```

IF([Products Below vs PM] > 0, "Green", "Red"))